D3.2 Inheritance (HL)

NAME	DATE	TOTAL
		_____ / 40

About 40 minutes, or one section at a time. The number of marks shows how many separate points to make. A chi-squared table is not needed: use 7.81 as the critical value for 3 degrees of freedom at p = 0.05.

Section A: Recall

8 MARKS

1 DEFINE linked genes. [D3.2.19] [1]

2 LIST the gametes made by a plant with genotype TtRr, where the genes are unlinked. [D3.2.16] [1]

3 STATE the expected phenotype ratio of a test cross between AaBb and aabb, where the genes are unlinked. [D3.2.17] [1]

4 DEFINE recombinant. [D3.2.20] [1]

5 STATE the stage of meiosis in which independent assortment happens. [D3.2.16] [1]

6 STATE how the degrees of freedom are calculated in a chi-squared test. [D3.2.21] [1]

7 DRAW the notation for an individual heterozygous for two linked genes, with A and B on the same chromosome. [D3.2.19] [2]

Section B: Understanding

14 MARKS

8 EXPLAIN how the behaviour of chromosomes in meiosis produces a 9:3:3:1 ratio in a dihybrid cross. [D3.2.16, D3.2.17] [4]

9 In tomatoes, red fruit (R) is dominant to yellow (r) and tall (T) is dominant to dwarf (t). The genes are unlinked. **PREDICT**, using a Punnett grid, the phenotype ratio from RrTt × Rrtt. [D3.2.17] [3]

10 **EXPLAIN** why linked genes do not assort independently. [D3.2.19] [2]

11 **EXPLAIN** how recombinants can arise from linked genes. [D3.2.20] [3]

12 The human genes HBB and HBD lie next to each other on chromosome 11. **PREDICT** how often their alleles would be separated by crossing over, and explain. [D3.2.18] [2]

Section C: Working with evidence

12 MARKS

All data in this section are invented for practice. The patterns are realistic.

13 [D3.2.20, D3.2.21]

In maize, coloured seeds (C) are dominant to colourless (c) and smooth seeds (S) are dominant to wrinkled (s). A plant heterozygous for both genes was crossed with a plant homozygous recessive for both. The offspring were:

PHENOTYPE	COLOURED, SMOOTH	COLOURLESS, WRINKLED	COLOURED, WRINKLED	COLOURLESS, SMOOTH	TOTAL
OBSERVED	168	162	36	34	400

a STATE the expected number in each class if the genes were unlinked. [1]

b STATE a null hypothesis for this test. [1]

c CALCULATE the chi-squared value. Show your working. [3]

d DEDUCE, with a reason, whether the null hypothesis should be rejected. [2]

e IDENTIFY the recombinant classes and calculate the percentage of recombinants. [2]

f DRAW the genotype of the heterozygous parent using linkage notation. [1]

g EXPLAIN why the recombinant classes are small. [2]

Section D: Longer answer

6 MARKS

14 **DISTINGUISH** between the results of test crosses involving two unlinked genes and two linked genes, explaining the differences. [D3.2.16, D3.2.19, D3.2.20] [6]